Lecture 5: Trend Following

Algorithmic Trading Course

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Outline

Trading Strategies

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A series of rules that implements buys and sells of a set of assets, given the information available just prior to the trade.

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where $dP_t \in \mathbf{R}^{k+1}$ a set of price changes, and $\theta_t \in \mathbf{R^{k+1}}$ is the weight-process (conditioned on the filtration $\mathcal{F}_t$).

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- In *discrete time* it is defined as

$$V(t_k) = V(t_0) + \sum_{1}^{k} \theta_k \cdot \Delta P_{k+1}$$

  where θ_k is known at time t_k, as is P_k and $\Delta P_{k+1} = P_{k+1} - P_k$.

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Trading Strategies (cont'd)

- We let P_t^0 be the money-market account, $dP_t^0 = r_t \; P_t^0 dt$ or $\Delta P_{k+1}^0 = r_k P_k^0$.

Dynamic Trading Strategies

A dynamic strategy is defined as a set of dynamically updated weights, also called the *signal* s_{t-1} multiplied by excess returns r_t, i.e., the strategy returns

$$r_t^s = s_{t-1} r_t$$

or, if we track final wealth

$$W_T = \sum_{t=1}^{T} r_t^s = \sum_{t=1}^{T} s_{t-1} r_t$$

The concept of strategy is used frequently in quantitative finance, in particular, *self-financing strategies*, see, e.g., Duffie, *Dynamic Asset Pricing Theory*, p 83ff.

Example signals

Strategy weights can be defined in a diverse number of ways, and we will focus on momentum strategies although other strategies may have similarly defined signals. (see, e.g, Firoozye-Koshiyama)

- Simple Moving Average (SMA):

$$s_{t-1} = \frac{1}{T} \sum_{1}^{T} r_{t-k}$$

- Exponentially-Weighted Moving Average (EWMA):

$$S_{t-1} = c(\lambda) \sum_{k=1}^{\infty} \lambda^k r_{t-k}$$

More example signals

- Difference between current price and moving average

$$s_{t-1} = p_{t-1} - \frac{1}{T}\sum_{1}^{T} p_{t-k}$$

- Differences between EWMAs:

$$s_{t-1} = c(\lambda_1)\sum \lambda_1^k r_{t-k} - c(\lambda_2)\sum \lambda_2^k r_{t-k}$$

- Differences between SMAs:

$$s_{t-1} = \frac{1}{T_1}\sum_{1}^{T_1} p_{t-k} - \frac{1}{T_2}\sum_{1}^{T_2} p_{t-j}$$

- (with no restriction on the signs of ϕ's this may not necessarily be a trend-following strategy) Forecasts from ARMA(p, q) models:

$$s_{t-1} = \phi_1 r_{t-1} + \ldots + \phi_p r_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q}$$

Final Comments on Signals

Note - Convolution kernel: All of these strategies can be characterised as $s_{t-1} = \sum_{k=1} \phi_k r_{t-k}$ for some set of weights (or *convolution kernel*) ϕ.

Strategies can also be non-linear functions of any of the above, e.g.,

$$s_t = 1_{\{\sum_1^T r_{T-k} > 0\}} - 1_{\{\sum_1^T r_{T-k} < 0\}}$$

going long one unit whenever the SMA of returns is positive, and short one unit whenever the SMA is negative.

We will focus on the simpler strategies which are linear in historic returns.

Example Strategies

Standard trading strategies which we will consider in various amounts of detail.

- Buy and hold
- Trend-Following - Time-Series
- Momentum - Cross-Sectional
- Mean-Reversion
- Carry
- Vol-based strategies (e.g., short-gamma)
- Others

Buy-and-Hold

Buy one unit of asset and hold over horizon.

- Boring!
- Only ever meant as a benchmark
- If Equities - you are long, so have a positive β to the market (usually).
- If Bonds - you are long bonds, which often are negatively correlated to Equities.
- 60-40 Strategy - 60% Equities + 40% bonds is a type of buy-and-hold strategy. Again a benchmark for a global macro fund.

Trend Following

Trend-Following

Momentum or Trend Following

- Most widely known Algo style
- Ben Graham (the father of *value investing* wrote about it (and invested in it)
- Proven to have worked for at least 200 years.

- Two major forms:
  - Cross-sectional, i.e., a Momentum factor
  - Time-series, e.g., a CTA
- Underlies almost all the CTA industry
- *Not* a diversified source of return
- Often thought of as an insurance trade (significant downside protection)

Defining Momentum

Definition (A version of Momentum)
In it's most basic form: Find a moving average/trend.

- Go Long if
  - Level $>$ Moving average (+threshold)
  - Short Moving Average $>$ Long Moving Average
  - Trend is going up
  - Some combination
- Go Short if
  - level $<$ Moving average (- threshold)
  - Short Moving Average $<$ Long Moving Average
  - Trend is going down
  - Some combination

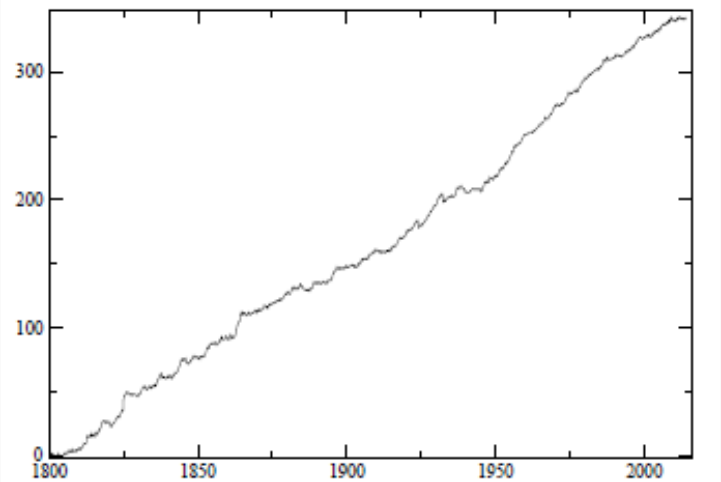
Example

Figure 1: USD-JPY and 15/30-day Moving Averages

Questions

- How do we establish trend?
- How far do we look back?
- Do we normalize signal?
- Do we cap and floor?
- Do we take other nonlinear transformations?

200 Years of Trend Following



sector	SR (T)	t-stat (T)	t-stat (T*)	SR (μ)	t-stat (μ)	start date
Currencies	0.47	2.9	2.9	0.1	0.63	1973
Commodities	0.28	4.1	3.1	0.3	4.5	1800
Bonds	0.4	3.9	2.7	-0.1	-1	1918
Indices	0.7	10.2	6.3	0.4	5.7	1800

Figure 2: Caption

Combined Sharpe Ratio of 0.72.

Lempérière et al - *Two Centuries of Trend Following* (2014)

Momentum Factoids

Momentum's desirable properties

- It tends to perform when vol is high. (counter: when vol is low it tends to underperform)
- It tends to have positively skewed returns (i.e., it may not win often, but when it wins, it wins big)
- The skewness of returns appears greater when measured over longer horizons (e.g., when investors look at quarterly returns they are very highly skewed, compared to daily returns).
- Its payoff is similar to that of a straddle, at least empirically. (Fung-Hsieh analysed this despite it being a heuristic)
- Makes money in trending markets, including down-trends (insurance)
- Loses money in sideways markets (pays for insurance)

We investigate the skewness over different horizons and We also investigate responsiveness to changes in trend.

Momentum and Skewness

Skewness Analysis

Skewness - Martin& Zou

- Let returns be $X_t \sim \mathcal{N}(0,1)$ IID
- We can still have a filter with nice properties
- Strategy returns $R_t = s_{t-1}X_t$ where s_{t-1} is signal
- Consider horizon T returns, i.e., $R_{t+T} = \sum_{\tau=t}^{T} s_{\tau-1}X_\tau$
- We consider EWMA signals

$$
\begin{aligned}
s_t &= \sum_{k=1}^{\infty} a_k X_{t-k} \\
&= c \sum_{k=1}^{\infty} \rho^k X_{t-k}
\end{aligned}
$$

where c is a normalizing constant

Calculating Skewness

Calculating Skewness

- We can take first three moments. Skewness is interesting
- Let innovation X_t be iid, with zero mean and 3rd moment and unit variance:
  - $\langle X_t, X_s \rangle = 0$ for $t \neq s$
  - $\langle X_t \rangle = 0$, $\langle X_t^2 \rangle = 1$ and $< X_t^3 >= 0$
- Zero mean and skewness
- Algebra alone will get that:
  - Mean:
    - $< R_{t+T} >= 0$
  - Variance:

$$
\begin{aligned}
\langle R_{t+T}^2 \rangle &= \langle (s_0 X_1 + s_1 X_2 + \cdots + s_{T-1} X_T)^2 \rangle \\
&= T \sum s_i^2 \langle X_{i+1}^2 \rangle \\
&= T \times \langle s^2 \rangle
\end{aligned}
$$

Skewness II

Skewness

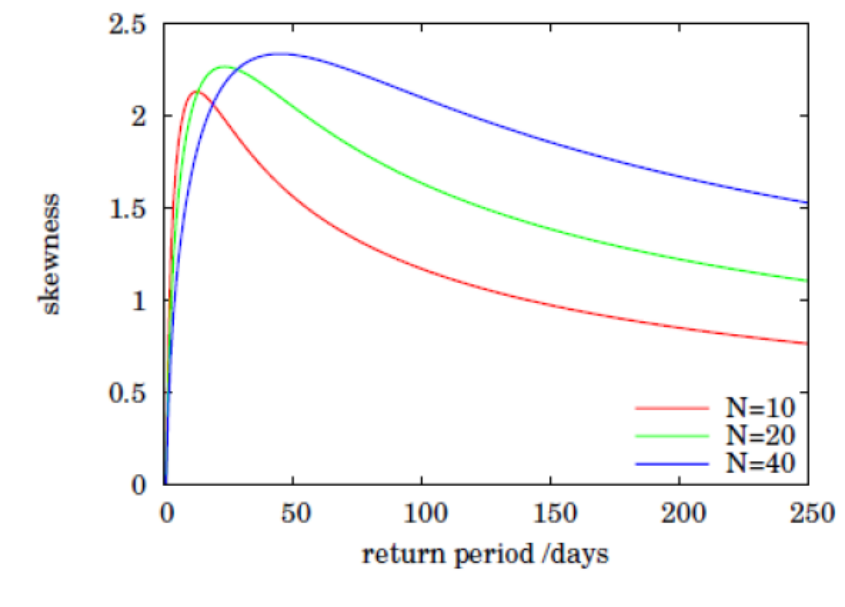
- 3rd Moment - Skewness:

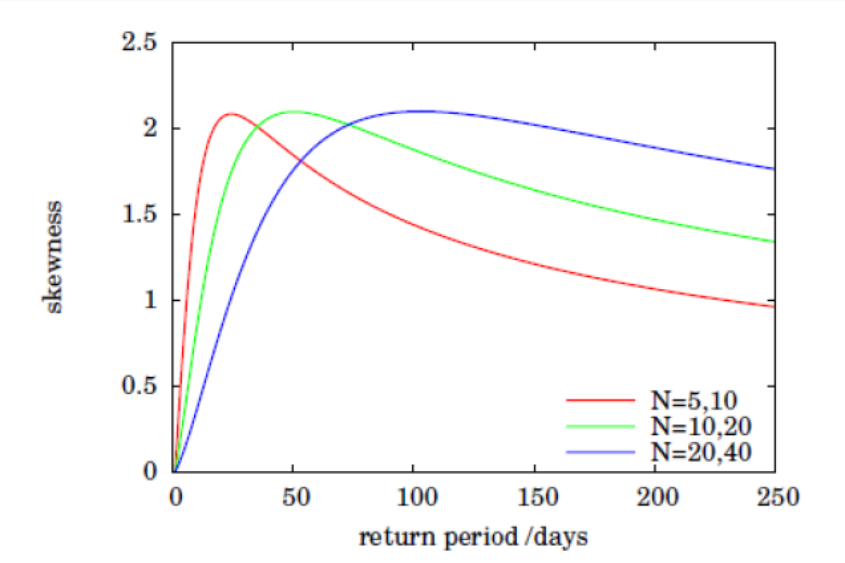
$$
\begin{aligned}
\langle R_{0+T}^3 \rangle &= \langle (s_0 X_1 + s_1 X_2 + \cdots + s_{T-1} X_T)^3 \rangle \\
&= \text{const} * \cancelto{0}{\langle s_{n_1} X_{n_1+1} s_{n_2} X_{n_2+1} s_{n_3} X_{n_3+1} \rangle} && \text{with } n_1 < n_2 < n_3 \\
&+ \text{const} * \cancelto{0}{\langle s_n^2 X_{n+1}^2 s_m X_{m+1} \rangle} && \text{with } n < m \\
&+ \text{const} * \cancelto{0}{\langle s_n^3 X_{n+1}^3 \rangle} && \text{with } n > 0 \\
&+ \text{const} * \langle s_n^2 X_{n+1}^2 s_m X_{m+1} \rangle && \text{with } n > m \\
&= 3 \sum_{0 \le n < m < T} \langle s_m^2 s_n X_{n+1} \rangle
\end{aligned}
$$

Since s_n^2 may depend on X_{m+1}

Skewness by Horizon

Skewness by horizon





Skewness III

Skewness
Now if X_k have a non-zero acf:

- Letting signal

$$s_k = \sum_{j=0}^{\infty} a_j X_{k-j}$$

  where a_j are fixed filter weights (e.g., $a_j = c \times \rho^j$ for EWMA)

- The third moment becomes

$$6 \sum_{1}^{T} (T - j) a_j \gamma_k(a)$$

- where γ_k is the autocovariance function of weights a i.e.,

$$\gamma_k^a = \sum_j a_j a_{j+k}$$

Fascinating skewness

Why is skewness so interesting?

- Even an un-skewed return distribution can lead to positive skewness
- (Almost all) risk-averse investors prefer:
  - more positive returns
  - less volatility
  - more positive skewness
- Why is skewness so strange?
  - Interaction between future signal (at time $T + k$) involving terms from T to $T + k - 1$
  - Interaction between future returns (at times T to $T + k - 1$)
- In reality mean, standard deviation and Sharpe also (even more) interesting!

Skewness and Risk Premia

Negatively Skewed have higher SR except for Momentum

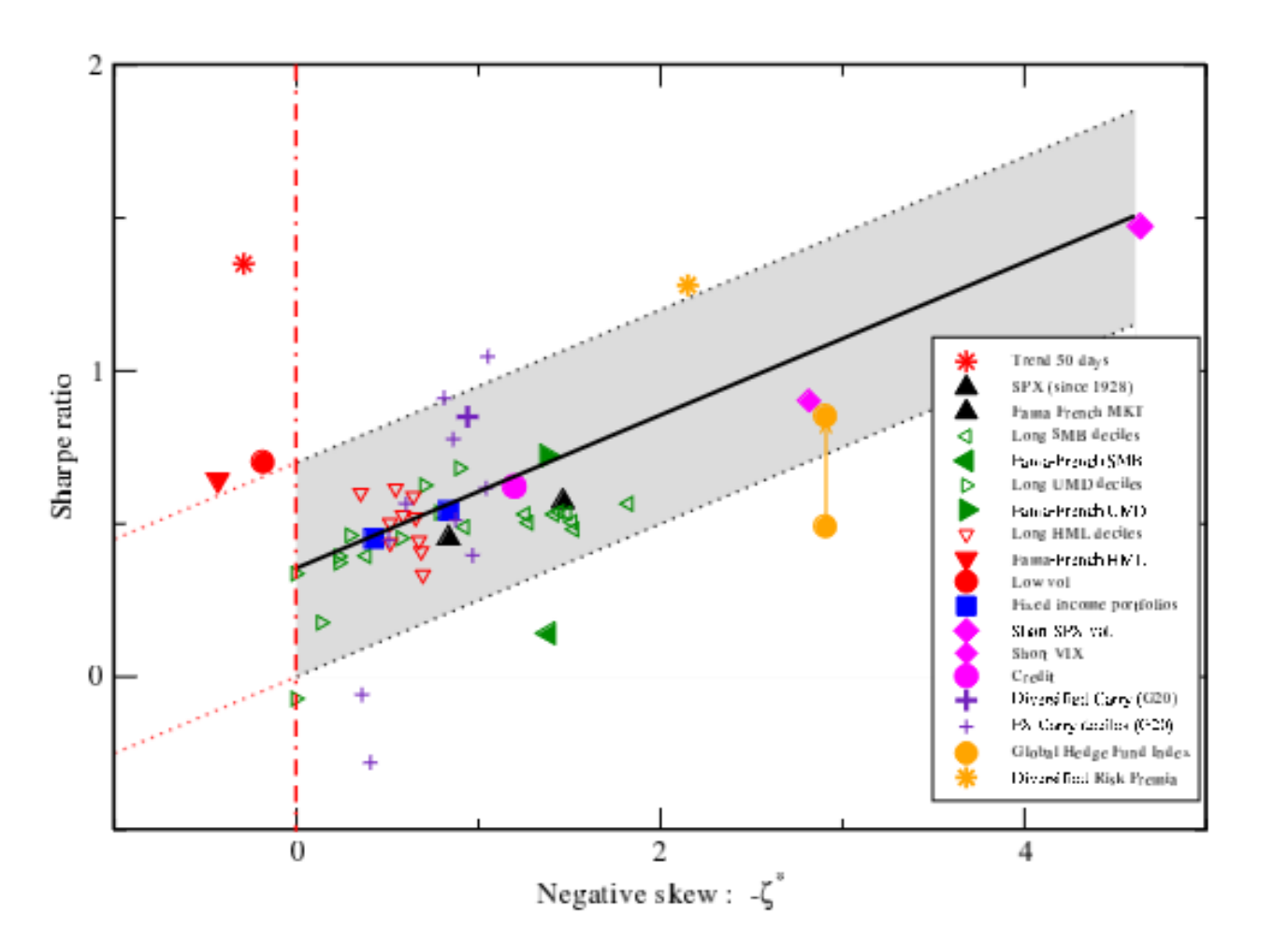


Figure 4: More negatively skewed have higher Sharpe Ratio. Source: Lemperiere et al, Risk Premia: Asymmetric Tail Risks and Excess Returns

Trend Following Practicalities

Many different methods all lead to reasonable trend-following strategies

- SMA, EWMA, combinations of signals (crossing EWMA, weighted EWMAs)
- Capped and floored signals (*Winsorising*)
- State-space models (local level and local slope models), etc.
- Loess, or local regressions
- ARIMA models
- Wavelets, Fourier decomposition

Taxonomy of filtering methods

Filtering returns

- Exponentially weighted moving averages (EWMA)
- Simple moving averages (SMA)
- Normalizing
- Winsorising
- Indicator Functions
- Combining Signals (Ensemble Methods)
- Nonlinear Transforms used as well

Taxonomy II

Filtering Levels (Differences of levels)

- Level vs EWMA
- Level vs SMA
- Crossing moving averages (specific order indicating momentum)
  - e.g., $\langle y \rangle_5 > \langle y \rangle_{10} > \langle y \rangle_{30}$ go very long,
  - $\langle y \rangle_5 < \langle y \rangle_{10} < \langle y \rangle_{30}$ go very short.
  - Sum or various horizon indicator functions of levels
- Differences of filters
- Holt-Winters (H-W) and local trend extraction

In spite of wide variety of models, momentum returns are almost identical
Levine and Pedersen, Which trend is your friend?

Skewness in Practice

Now, despite the fact that the time series of returns is here assumed to be a sequence of iid variables following a normal distribution, the strategy returns r^s_{t+T} time series has different properties, in particular it displays skewness, i.e. trend followers lose more often than they gain, as shown in a paper by Bouchaud and Potters (2005).

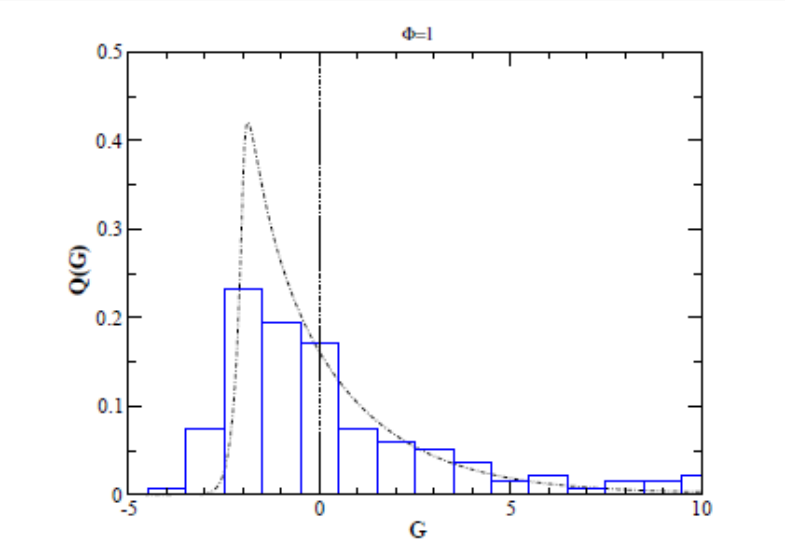


Figure 5: Caption

Skewness and Nonlinear Transforms

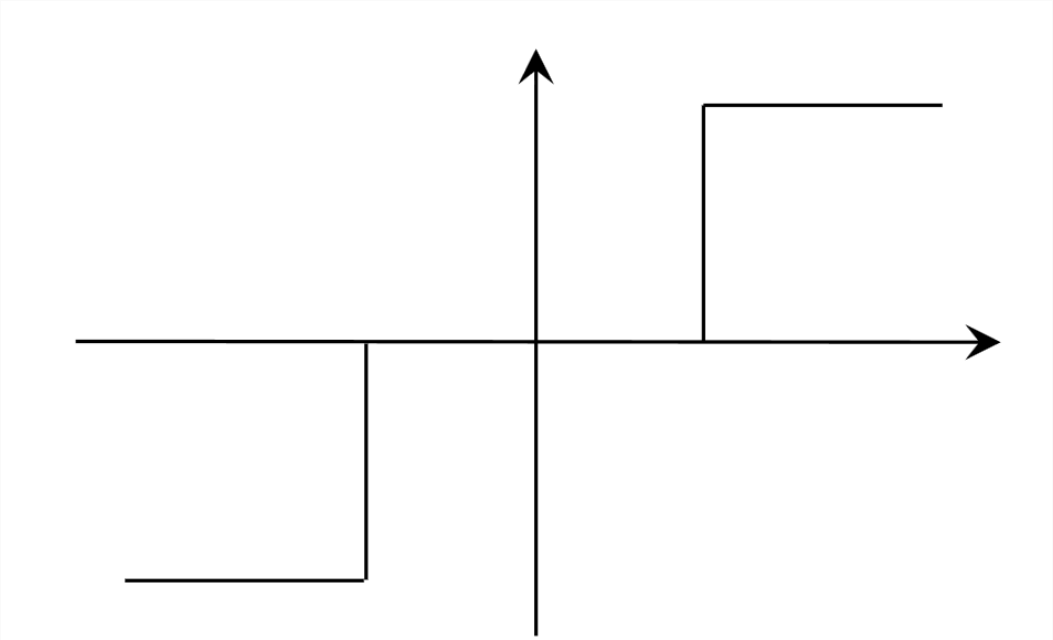


Figure 6: Step Function

Skewness and Nonlinear Transforms

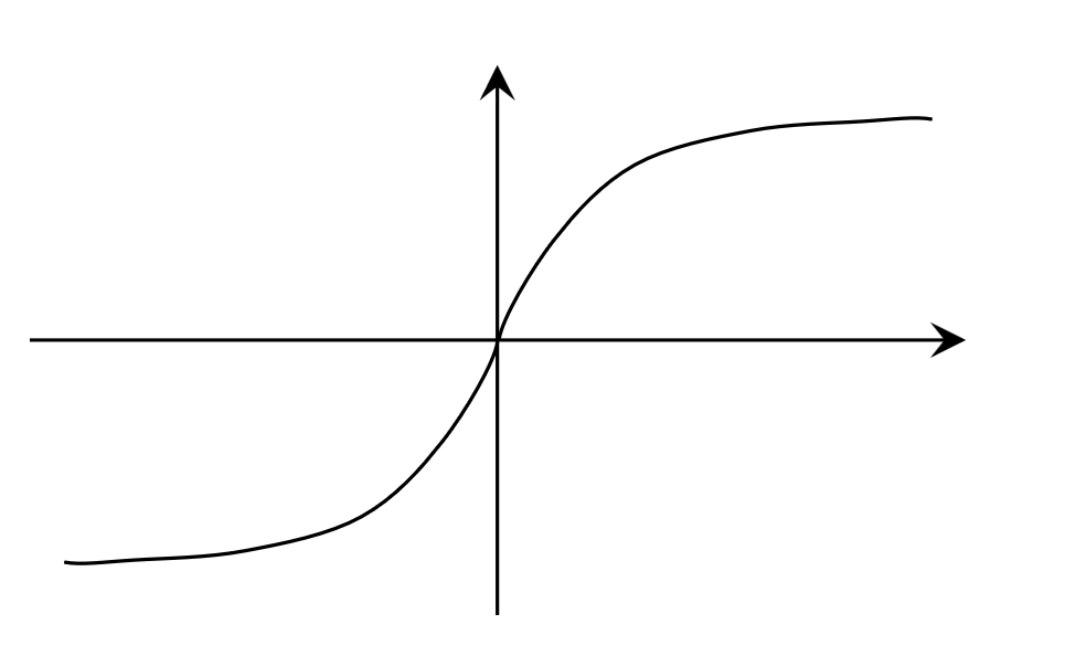


Figure 7: Sigmoid

Skewness and Nonlinear Transforms

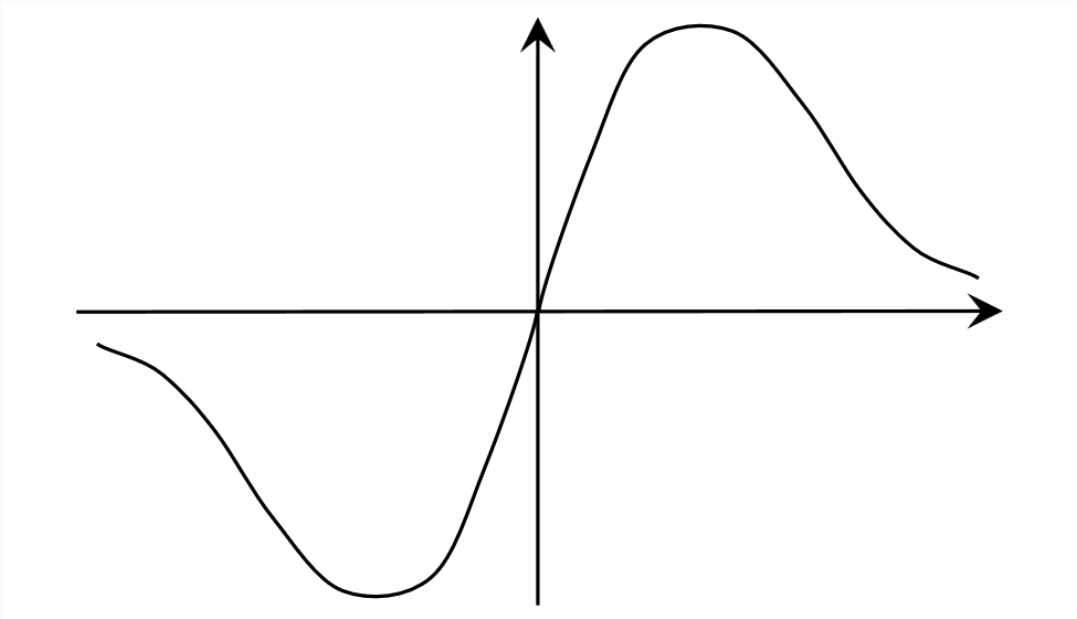


Figure 8: Reverse Sigmoid

Skewness and Nonlinear Transforms

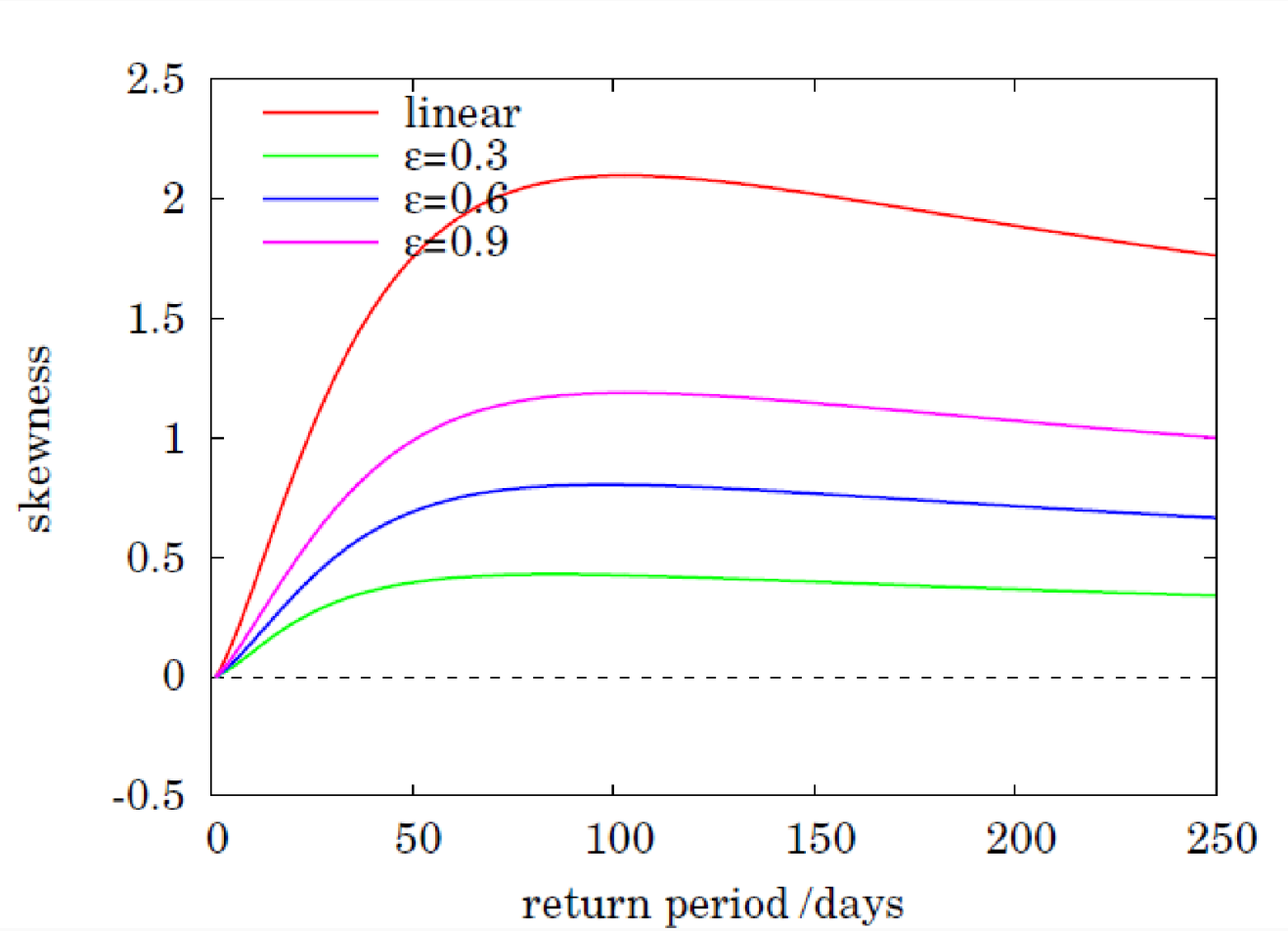


Figure 9: Step Skewness - threshold lowers it

Skewness and Nonlinear Transforms

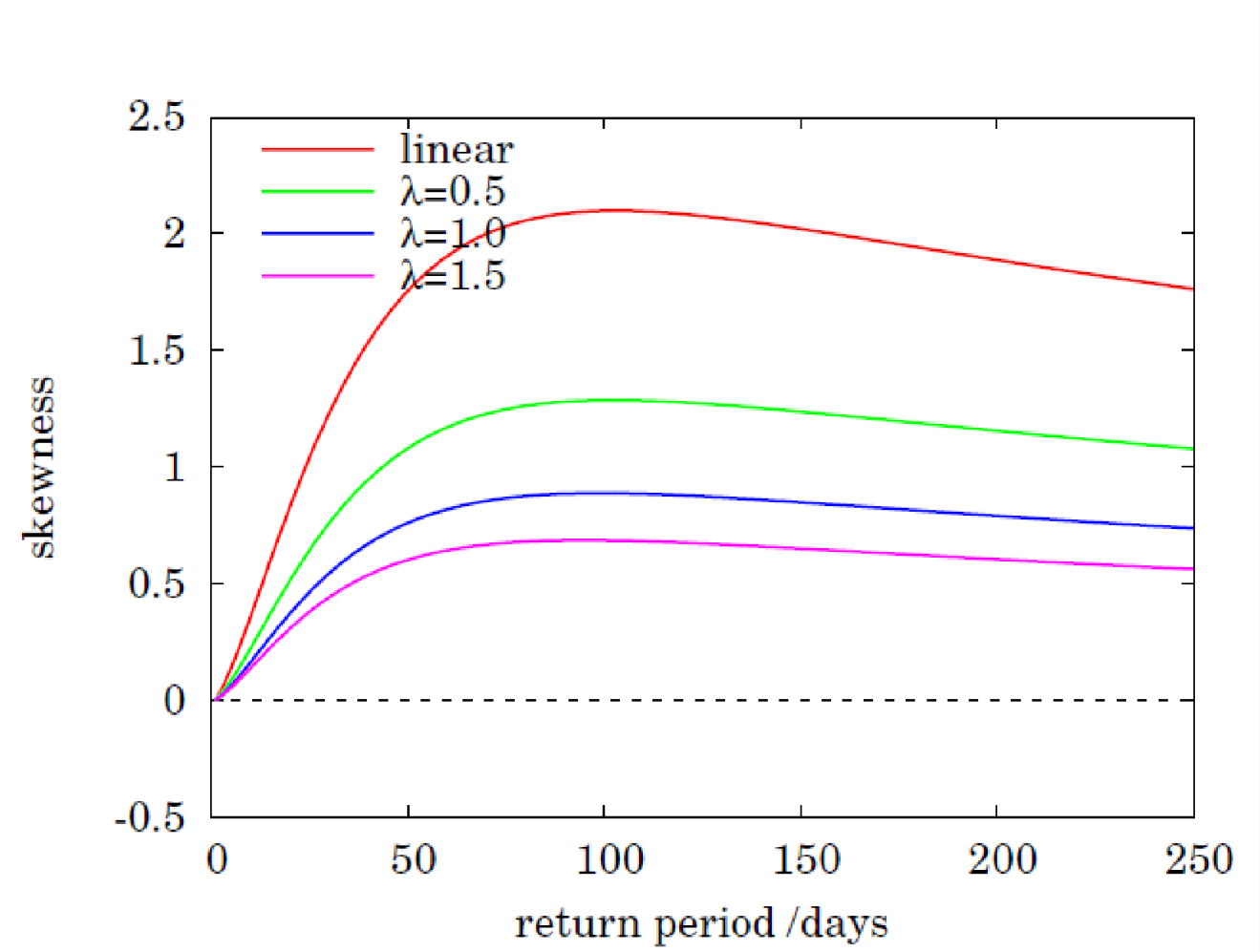


Figure 10: Sigmoid Skewness - peak closer in lowers it

Skewness and Nonlinear Transforms

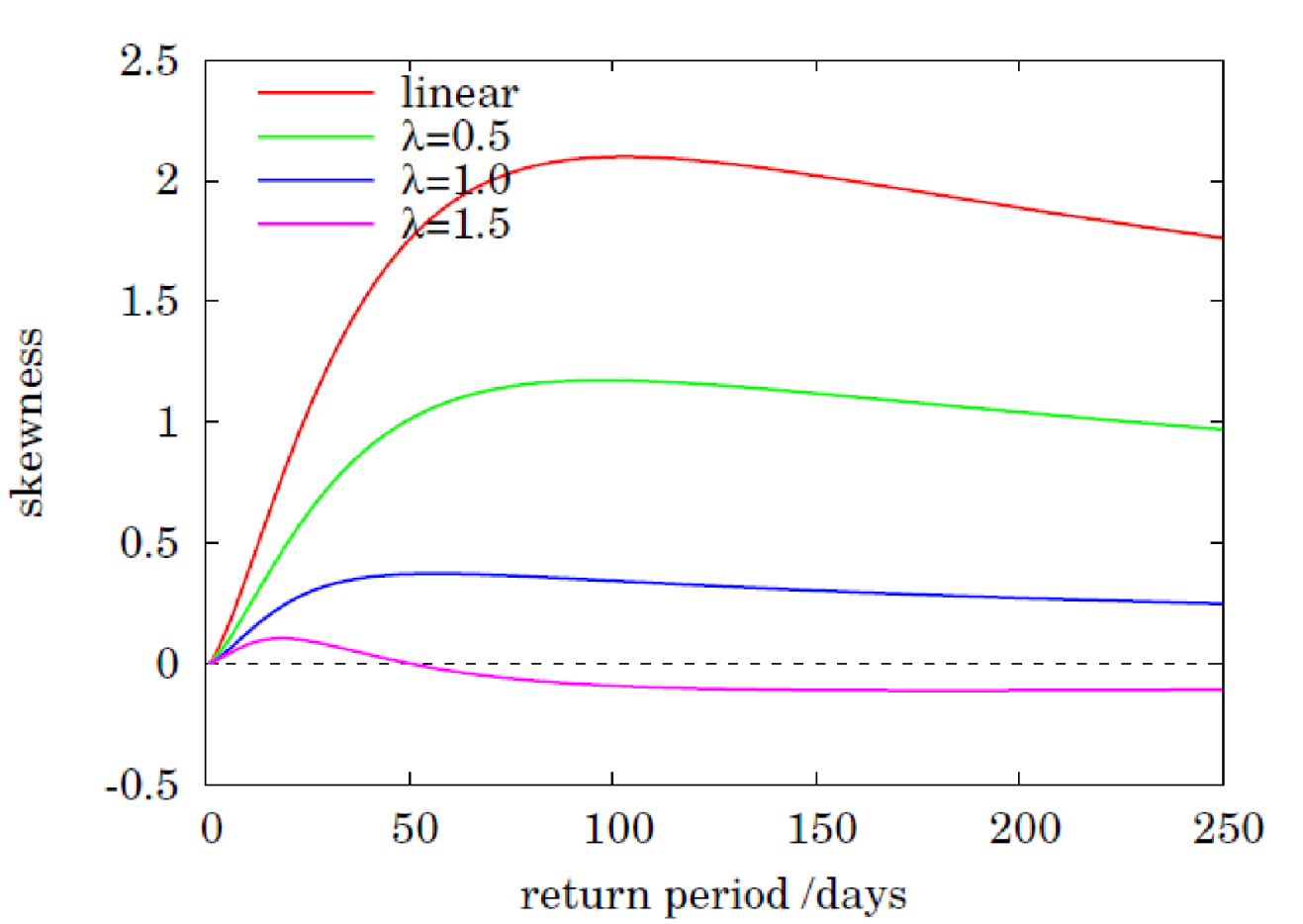


Figure 11: Reverse Sigmoid Skewness - peak closer/falls quicker in lowers it

Momentum and Responsiveness

Responsiveness

- Short-term MA -
  - more responsive - captures established trends near their start
  - higher transaction costs - bleeds away in a sideways market
- Long-term MA
  - less responsive - lagged response to new turning points
  - lower transaction costs - doesn't lose money in a sideways market

Trend via SDEs

Trend via SDE

- Consider a nonlinear allocation depending on levels $\theta_t = f(S_t)$
- The total PnL is then $dV_t = f(S_t)dS_t$.
- Let $F(S) = \int_c^S f(s)ds$, then, if S is deterministic

$$\begin{aligned} V(T) - V(0) &= \int_0^T f(S_t)dS_t \\ &= F(S_T) - F(S_0) \end{aligned}$$

- If S is stochastic, $dS = \mu_t S_t dt + \sigma_t S_t dW$ by Ito, we have

$$V(T) - V(0) = \underbrace{F(S_T) - F(S_0)}_{\text{option profile}} - \underbrace{\frac{1}{2}\int_0^T f'(S_t)S_t^2\sigma_t^2 dt}_{\text{trading impact}}$$

Trend via SDEs

Trend via SDE

We let

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_t dW$$

and define

$$d\hat{\mu}_t = -\lambda \hat{\mu}_t dt + \lambda \frac{dS_t}{S_t}$$

$\hat{\mu}$ is an EWMA, i.e.

$$\hat{\mu}_t = \int_{-\infty}^{t} e^{-\lambda(t-s)} \frac{dS_s}{S_s}$$

Note that we still cannot represent all ARMA processes using SDE. Only MA terms. If we consider the trend following strategy using a single EWMA:

$$\frac{dV_t}{V_t} = m \frac{\hat{\mu}_t}{\sigma^2} \frac{dS_t}{S_t}$$

SDE approach Cont'd

Then we have the following identity (just using Ito's Lemma)

$$\ln\left(\frac{V(T)}{V(0)}\right) = \underbrace{\frac{m}{2\lambda}(\hat{\mu}_T^2 - \hat{\mu}_0^2)}_{\text{option profile}} + \underbrace{m\sigma^2 \int_0^T \left(\frac{\hat{\mu}^2}{\sigma^2}\left(1 - \frac{m\sigma^2}{2}\right) - \frac{\lambda}{2}\right) dt}_{\text{trading impact}}$$

- **Option profile** evolves quickly, lower bound of $-\frac{m}{2\lambda}\hat{\mu}_0^2$, and thus is more exposed to risky asset if the initial trend is high (at the start).
- **Trading impact** cumulative, evolves slowly. $m \ll 1$ then there is a positive contribution whenever

$$|SR| > \sqrt{\frac{\lambda}{2}}$$

SDE approach

SDE approach Cont'd

A few sources that specifically use the SDE approach:

- Bruder-Gaussel: Risk-Return Analysis of Dynamic Investment Strategies
- Jusselin et al: Understanding the Momentum Risk Premium
- Hamdan et al: Alternative Risk Premia

Cross-Sectional Momentum

Cross-Sectional

Cross-Sectional Momentum

- We will cover this in some detail later on in factor investing.
- Cross Sectional Momentum: Buy the winners, sell the losers.
- Carhart, On Persistence of Mutual Fund Performance was the first to introduce into Equities
- Usually 12-month (less final month) momentum, ranking equities. Go long top-decile, short bottom-decile.
- It is one of the most significant Equities factors
- Arguably, this is one of many methods for forming a portfolio.

Cross-Sectional vs Time-Series

Cross-Sectional vs Time-Series

- Continued work on relationship.
- Some indication that Time-Series momentum works better.
- Time-series is only concerned with winners when they are outperforming or losers when they are underperforming
- Cross-sectional is concerned with relative performance. Digs deeper, including some stocks that may not be doing well in absolute terms
- Time-Series may be net-long or net-short as a consequence. Cross-sectional can be market-neutral.
- This is an active area of research.

Ranking in Portfolios

Cross-Sectional Momentum and Ranking Portfolios

- Long the top decile and short the bottom decile, cash neutral is just one possible portfolio.
- There is an entire topic on portfolios from sorts.
- Almgren-Chriss, Portfolios from Sorts
- Nguyen-Lo Robust ranking and portfolio optimization
- Often the ranking of assets should have some difference.
- This is an active area of research.

Other Topics in Momentum

Other Topics in Momentum

Mean-Reversion, Momentum, Mean-Reversion

Almost all assets seem to

- Mean Revert over very short horizons (at various time-scales, intraday and over a few days)
- Trend over the medium Horizon
- Mean-revert (or have reversion to long-term value) over the very long-horizon
- Changes to this behaviour are possible. And certainly breaks/break-outs/change-points are a profitable area,
- In general, if we keep the horizon the same each of these strategies can be profitable

Mean-Reversion, Momentum, Mean-Reversion

Figure 12: US Treasury Yields

Momentum - Over Utilised?

Momentum Crowding

- Momentum funds have grown rapidly over past 15 years.
- There is some cause to believe that returns are lower because of crowding
- In fact, Winton argued that some CTA AUMs were not CTA funds (from our first day's lecture) , primarily because this was a touchy point.
- Man AHL, recognising challenges, launched *AHL Evolution* in 2006, to take advantage of momentum in *less liquid* assets (funnily enough - 2nd or longer utures contract, swap forwards, bonds etc)

Trading Futures

Information on Futures Markets

No one source – Combine several

Information on Futures Markets

- Bloomberg (WEIF, WBF, Comdty pages, etc)
- Tick Data
- CSI Data
- Pure Financial Academy
- MRCI
- Bar Chart

Bloomberg for Futures

Futures DES pages:

- Exchange
- Underlying
- Contract Size
- Tick Size
- Tick Value
- Calendar

Other FLDS:

- EDA Comdty FLDS ¡go¿ Size ¡go¿
- NGA Comdty FLDS ¡go¿
- Underlying ¡go¿

Futures research sources

Tickdata - https://www.tickdata.com

- Paid service Rolls properly

CSIdata - http://www.csidata.com

- Lots of info
- Funny symbol names

Barchart - https://www.barchart.com

- Many other relevant details

DataBento - https://databento.com

- Good deal of futures data, including LOB, very reasonably priced.

Other sources

- Good and relevant information (sometimes spotty coverage)
- Inconsistent naming conventions
- Have to merge all with manual adjustments
- Double-check and verify

Futures Master Table

This is always a work in progress, but at minimum contains:

Futures Master Table

- Ticker
- BBGType - Index, Comdty
- Exchange
- Bloomberg code
- Holidays
- Description of underlying
- Currency
- Delivery Type – Cash or Physical
- Contract Size
- Delivery Months
- Tick Value

Rolling Futures

Rolling Futures

- Futures must be rolled to avoid cash settlement/physical settlement
- Roll dates based on either:
  - Min(last trade date, first notice date, first delivery date) – k days
  - When the volume of the 2nd exceeds that of the 1st contract
  - Can be rolled proportionally (reinvesting for a different number of contracts) or by shifting (keeping contract numbers the same).
- Need the key dates and volumes for first 2 contracts. Can roll some futures sporadically rather than deterministically.
- Roll entire chain at same time – 1st, 2nd, etc, out to Nth contract.
- Rolling incurs one-sided transaction costs.
- References
  - Continuous Futures Contracts for Backtesting
  - Continuous Futures Methodology for Backtesting
  - Cont Futures for Backtesting and Technical Analysis

Futures Expiry Tables

Futures Expiry Tables

- Contract name for each contract
- Expiry Month
- Last Trade
- First Notice
- First Delivery
- Last Delivery
- Last Price

Futures Naming Convention

Futures Naming Conventions

Contract names usually 2/3 letter prefix:

- ED = Eurodollar
- ES = SPX mini
- TY = 10y-UST bond
- G = Gilt (with space)
- SFFR = SOFR

Next comes delivery month:

- Monthlies: Jan:F, Feb:G, Mar:H, Apr: J, May: K, Jun:M, Jul:N, Aug:Q, Sep:U, Oct:V, Nov:X, Dec:Z
- H, M, U, Z = quarterlies, Others = Serials

Finally comes year:

- Usually 1 digit, but BBG has schedule to convert to 2 digits

Updating Futures Data

Storing Futures Data

- Periodic bulk download of all the timeseries needed OR use a real-time downloader which updates all series on fixed schedule.
- Make sure full history of expiries are available (for proper rolling).
- Bulk download missing series etc.
- Pull in reference index if available (for carry calcs, signal)
- Convert all to at least 2-digit year expiry (if not 4-digit)
- Store in a suitable database.
- Must store unadjusted futures prices (for levels and carry calcs)

Other Considerations

Other Considerations

- Can store intraday (eg, BBG has HLOC ASK, BID and MID at 1-min or longer. Refinitiv has long histories).
- Store opening times. (see e.g., Pandas Market Calendars)
- Close may not coincide with pricing time for deliverable. (what is the carry of the first contract?)
- Closing = Last = Final Trade
- Settle Price - Where traders will mark mid- How is it determined?
- Bid / Ask - sticky or real?

An Exercise

Exercise

Background - Momentum in Equities

- Recall - Equities Indices typically trend in the medium term (e.g., 1y), but mean-revert in the short term (e.g., 10 days)
- Carhart recognised this and created the momentum factor sorting returns from 12m less the previous month.
- At that point buy the winners and hold.
- It is also known that momentum stops working after about 18m. Sometimes this is called mean-reversion even though the term is loose. (It is also unclear if we need to worry about this)

Exercise

Exercise - 1

- Download an equities index (e.g., SPX Index, SX5E Index, NKY, CAC, HSI etc) or futures series (CL1, CO1, NG1, GC1, C 1, etc), or ETF (SPDR, etc). At least 5 years.
- If a futures, make sure you account for the roll properly - 2-3 days before first notice date.
- Download an approx risk-free rate (SOFR - repo rate, or Fed Funds or SONIA or similar)
- Note: First Notice ¿ First Delivery ¿ Last Delivery $\approx$ Last Trade Date (Discussed in Futures section)

Exercise

Exercise - 1b

- Preferable to account for bid/ask, i.e., forecast and mark using

$$\text{mid} = \frac{\text{bid} + \text{ask}}{2}$$

but enter and exit incurring costs

$$\text{tc} = \frac{\text{ask} - \text{bid}}{2}$$

Exercise-2

Excercise - 2

- Create an appropriate momentum signal given you know what you know.
- Save at least 1y for testing
- "Train" by adapting for filter parameter(s) to ensure decent performance on the test set.
- By performance we mean Sharpe Ratio (Avg Excess Return/Stdev Returns). Do it by year and for entire train set. You can include $ Returns or % returns, etc. Adjust as you wish. Only when you are done, apply to test set.
- How have you accounted for what you know about the time-series?

References

References

Time-series and cross-sectional momentum

- Asness, Moskowitz and Pedersen, "Value and momentum everywhere." The Journal of Finance 68.3 (2013): 929-985.] Momentum in all asset classes
- Moskowitz-Ooi-Pedersen, Time-Series Momentum
- Lecture Notes Moskowitz' Lecture Notes Documents significant momentum in Equity index, currency, commodity, and bond futures for 58 instruments

References

Varieties of Trend-following methods

- Bruder, Zhao, Richard, Roncali, Trend Filtering Mechanisms for Momentum Strategies Practicalities of filtering - one method for trend-following
- Moskowitz and Pedersen Which trend is your friend? Present a variety of different trend method, Many different methods can be used to construct basically the same signals

References

Other papers of note

- Fung-Hsieh - first presented empirical paper on straddle-like payoff Fung Hsieh Risk in HF strategies
  - Some important empirical work on CTAs, showing the straddle-like payout.
  - The theory is very facile (daily lookback straddles do *not* and *never did* exist).
  - The strategies are never perfectly efficient as would be true of the exotic lookback straddle.
- Lo-Kaminski When do stop losses stop losses?Also quite simple paper, Stop losses work only for trending series, hurt for mean-reverting
- Firoozye-Kohiyama Avoiding Backtest Overfitting - Describes dynamic strategies including momentum and mean-reversion and 'optimal' ways of choosing how to weight them.

What have we covered

Learnings

What we covered

- Trading Strategies - Definitions
- Momentum, its history and pervasiveness
- Cross Sectional and Time-series Trend
- Momentum and Skewness
- Momentum's Responsiveness
- Futures, How to store them, how to roll them
- An Exercise in Momentum trading